

Reggie Smith

San Diego, CA

rjsmit13@gmail.com | (858) 212-2820

GitHub: <https://github.com/sreg23>, <https://sreg23.github.io/> (Portfolio)

LinkedIn: <https://www.linkedin.com/in/reggie-smith-03003b10a/>

EDUCATION

Oregon State University

B.S. Computer Science - Completed Spring 2026

Santa Monica College, CA

A.S. Computer programming - Completed Fall 2020

Relevant Coursework:

Data Structures, Algorithms, Operating Systems, Networking, Databases, Software Testing, Cybersecurity, Computer Graphics

TECHNICAL SKILLS

Languages

Python, C, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Technologies

React • Vite • Node.js • Express • REST APIs • JWT Authentication • bcrypt • MySQL • MongoDB • Docker

Cloud & Deployment

Render • Aiven Cloud Database

Tools

Git • GitHub • Linux • VS Code • MySQL Workbench • PyCharm

Concepts

Data Structures • Algorithms • Relational Database Design • Authentication & Authorization • Web Design • Networking • Socket Programming • Software Testing • Operating Systems • Cybersecurity

TECHNICAL PROJECTS

Job Tracker Web Application

- Built full-stack job tracking application using Node.js, TypeScript, Express, MySQL, and Vite + REST APIs
- Designed relational database schema supporting many-to-many relationships and user authentication
- Implemented CRUD operations for jobs, contacts, and skills management
- Collaborated in a GitHub-based team development workflow using pull requests and version control

OTP Client/Server Encryption System

Networking Project | C, Linux Sockets

- Developed concurrent TCP client/server encryption system using Linux socket programming
- Implemented one-time-pad encryption and secure client/server communication workflows
- Built custom handshake validation and process-based concurrency mechanisms
- Designed network protocol logic for encrypted message transmission

Reliable Data Transfer Protocol

Networking Project | Python

- Implemented reliable data transfer protocol over an unreliable network environment
- Developed timeout handling, retransmission logic, acknowledgments, and flow control mechanisms
- Analyzed packet delivery reliability and protocol performance under simulated network failures

Custom Unix Shell (smallsh)

Operating Systems Project | C, Linux

- Developed a custom Unix shell supporting command execution, input/output redirection, background processes, and signal handling
- Implemented Linux process management using `fork()`, `exec()`, `waitpid()`, and signal APIs
- Replicated core Bash functionality using low-level operating system concepts

Bloom Filter Password Detection Project

- Implemented Bloom Filter in Python for probabilistic weak password detection
 - Evaluated false-positive behavior and runtime performance using large password datasets
 - Analyzed tradeoffs between memory efficiency and detection accuracy
-

EXPERIENCE

GenMark Diagnostics — Manufacturing Lab Team Lead

Oceanside, CA | Dec 2016 – Dec 2018

- Led manufacturing operations in regulated FDA/ISO production environments
- Analyzed process metrics and implemented corrective actions to improve operational performance
- Collaborated cross-functionally with Engineering, R&D, QA, Regulatory, and Technical Operations teams
- Developed training plans and supported process documentation and compliance activities

GenMark Diagnostics — Manufacturing Lab Technician

Oceanside, CA | Nov 2015 – Dec 2016

- Supported production of automated molecular diagnostic products in clean-room environments
- Assisted with process improvement initiatives, SOP development, and quality investigations
- Applied lean manufacturing principles to improve efficiency and reduce cycle time

Ultimate Labs, Inc. — Operations Specialist

San Diego, CA | Mar 2013 – May 2015

- Managed laboratory operations and client compliance activities
- Performed data analysis, quality investigations, and environmental monitoring
- Coordinated testing workflows and reporting activities across multiple projects

Independent Contractor

San Diego, CA | Mar 2020 – Present

- Web Design
- Warehouse technician
- Delivery services

Additional Academic Experience

- Applied black-box, white-box, unit, and randomized testing methodologies in software engineering coursework
- Implemented CI-based development workflows using GitHub pull requests and automated testing
- Completed projects involving cybersecurity, network protocols, operating systems, database systems, and full-stack web development